



VIDYAPEETH

IIT-JEE | NEET | FOUNDATION

11th
JEE Main

ROI + KPM

Milestone Test-01

Phase-05

DURATION: 180 Minutes

DATE: 13/09/2026

M.MARKS: 300

Topic Covered

Physics : Mathematical Tools
Chemistry : Structure of Atom
Mathematics : Basic Maths Part Test - Number system, Wavy Curve Method

GENERAL INSTRUCTION

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Mathematics**) and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions

1. Use blue/black dark ballpoint pens.
2. Darken the bubble completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be ready by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple marking will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**
8. In integer answer type if the answer is 590 then fill it in OMR as 5 in Hundreds place (H row), 9 in Tens place (T row) and 0 in Ones place (O row).

EXAMPLE: H ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9
T ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☒ 9
O ☒ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

Name of the Student (In CAPITALS) :

Roll Number : _____

OMR Bar Code Number : _____

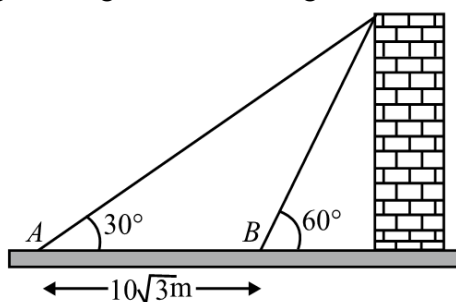
Candidate's Signature

Invigilator's Signature

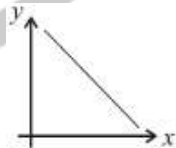
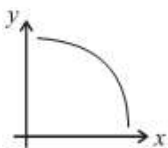
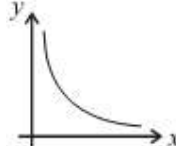
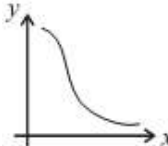
SECTION-I (PHYSICS)

Single Correct Type Questions (1-20)

1. Find approximate value of the $\sqrt{0.95}$:
 (1) 1 (2) 0.595
 (3) 0.60 (4) 0.975
2. Angle of elevation is the angle which line of sight makes with the horizontal. Angle of elevation of the top of a tall building is 30° from a place A and becomes 60° from another place B that is $10\sqrt{3}$ m from A towards the building as shown in the figure. Height of the building is close to



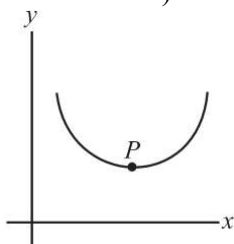
- (1) 7.5 m (2) 10 m
 (3) 12.5 m (4) 15 m
3. The equation of the circle passing through (4, 5) having the centre at (2, 2) is:
 (Standard form of circle is $(x - x_1)^2 + (y - y_1)^2 = R^2$, where x_1 and y_1 are coordinate of centre and R is radius of circle)
 (1) $x^2 + y^2 + 4x + 4y - 5 = 0$
 (2) $x^2 + y^2 - 4x - 4y - 5 = 0$
 (3) $x^2 + y^2 - 4x - 13 = 0$
 (4) $x^2 + y^2 + 4x - 4y + 5 = 0$
4. Find the value of $(1 + x)^3$, if $x \ll 1$.
 (1) $1 + x$ (2) $1 - 3x$
 (3) $1 + 3x$ (4) $1 + 3x + 3x^2 + x^3$
5. The equation $\sqrt{x} = 2y$, represents that graph between x and y is a :
 (1) Straight line (2) Parabola
 (3) Hyperbola (4) Circle
6. Among all straight lines through $P(1,2)$ meeting the line $2x - y + 1 = 0$, the minimum possible length of the segment intercepted between P and the meeting point is
 (1) $\frac{1}{\sqrt{5}}$ (2) $\frac{2}{\sqrt{5}}$
 (3) $\frac{3}{\sqrt{5}}$ (4) $\sqrt{5}$

7. $\log_e 8$ is equal to
 (1) $\log_e 2$ (2) $2\log_e 2$
 (3) $3\log_e 2$ (4) $4\log_e 2$
8. The tangent to the curve, $y = 2 \ln x + 2$ at the point (1, 2) makes an angle θ with the positive x -axis. Then
 (1) $\tan \theta = 4$ (2) $\tan \theta = 1$
 (3) $\tan \theta = 3$ (4) $\tan \theta = 2$
9. If $y = (\sin x)^2$ then find $\frac{dy}{dx}$:
 (1) $2 \sin x$ (2) $2 \cos x$
 (3) $2 \sin x \cdot \cos x$ (4) $2 \cos^2 x$
10. $\log 25 + \log 4 - \log 5$ is equal to
 (1) $\log 20$ (2) $\log 25$
 (3) $\log 15$ (4) $\log 10$
11. What is the value of $\log_5 512$, when $\log 2 = 0.3010$ and $\log 3 = 0.4771$?
 (1) 3.912 (2) 3.876
 (3) 2.967 (4) 2.896
12. The value of $\sin \frac{\sin 60^\circ}{\cos 45^\circ}$ is
 (1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{\sqrt{2}}$
 (3) $\sqrt{\frac{3}{2}}$ (4) $\sqrt{\frac{2}{3}}$
13. For $x > 0$, graph of $x^2 y = 2$ is best represented by:
 (1)  (2) 
 (3)  (4) 
14. Find the sum of first 50 natural numbers.
 (1) 637.5 (2) 1275
 (3) 1225 (4) 612.5
15. A periodic function passes through the origin, reaches a maximum at $x = 90^\circ$, and returns to zero at $x = 180^\circ$. What function does this graph represent?
 (1) $y = \cos x$ (2) $y = \sin x$
 (3) $y = \tan x$ (4) $y = \cos 2x$

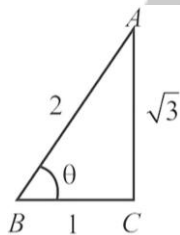
16. Find the sum of the infinite series:

$$Q = 6 + 2 + 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

- (1) 9 (2) 10
(3) 10.5 (4) 11
17. What is value of expression $2(\sin 15^\circ + \sin 75^\circ)^2$?
(1) $3/2$ (2) $1/2$
(3) 2 (4) 3
18. At point P , the value of slope is: (P is the lowest point of the smooth curve)



- (1) Zero (2) Positive
(3) Negative (4) Infinite
19. Find the value of angle $\angle ABC$.



- (1) 0° (2) 60°
(3) 30° (4) 45°

20. The square root of $(7 + 3\sqrt{5})(7 - 3\sqrt{5})$

- (1) 4 (2) $\sqrt{5}$
(3) $3\sqrt{5}$ (4) 2

Integer Type Questions (21-25)

21. If α and β are roots of the equation $10x^2 - 25x + 5 = 0$. The value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is:
22. If $\log_x \log_{18}(\sqrt{2} + \sqrt{8}) = \frac{1}{3}$, then the value of $32x$
23. 3, 6, 9, 12, 15, 120. The sum of series is x . Find $\frac{x}{10}$.
24. Find sum of first ten terms of given Arithmetic progression: $1 + 3 + 5 + 7 \dots$ upto 10 terms.
25. Find the sum from the sixth term to the twelfth term of the arithmetic progression 6, 10, 14,

SECTION-II (CHEMISTRY)

Single Correct Type Questions (26-45)

26. Out of the following types of electromagnetic radiation, which one has the highest frequency?
(1) X-rays (2) Ultraviolet rays
(3) Microwaves (4) Visible light
27. When the emission spectrum of H-atom during transition from energy level 'n' to ground state in one or more steps is observed, line belonging to Brackett series is not obtained. The option which correctly represents the wave number of lines belonging to Balmer series may be
(1) $\frac{8R}{9}, \frac{5R}{36}$ (2) $\frac{3R}{16}, \frac{8R}{9}$
(3) $\frac{5R}{36}, \frac{3R}{16}$ (4) $\frac{3R}{4}, \frac{3R}{16}$
28. The speed of electron revolving in the fourth orbit of a hydrogen-like atom or ion is 1094 km/s. The atom or ion is:
(1) H (2) He^+
(3) Li^{2+} (4) Be^{3+}

29. Which one of the following sets represents a collection of isotones?
(1) $^{23}_{11}\text{Na}^+$, $^{40}_{20}\text{Ca}^{2+}$, $^{24}_{12}\text{Mg}^{2+}$
(2) $^{14}_6\text{C}$, $^{16}_8\text{O}$, $^{15}_7\text{N}$
(3) ^9_4Be , $^{27}_{13}\text{Al}^{3+}$, $^{35}_{17}\text{Cl}^-$
(4) $^{40}_{20}\text{Ca}^{2+}$, $^{133}_{55}\text{Cs}^+$, $^{80}_{35}\text{Br}$
30. Among the following representations of ground states of atoms which is impossible?
(1) $1s^2 2s^1$ (2) $[\text{Ne}]3s^2 3p^6 4s^1$
(3) $1s^2 2s^2 2p^6 3s^2$ (4) $[\text{Ne}]3s^2 3p^6 4s^3$
31. An electron is moving in 3rd orbit of Li^{+2} and its separation energy is y . The separation energy of an electron moving in 2nd orbit of He^+ is
(1) $\frac{4y}{9}$ (2) $\frac{y}{9}$
(3) $\frac{-y}{9}$ (4) y

32. From the α -particle scattering experiment, Rutherford concluded that:
- α -particles can come within a distance of the order of 10^{-14} m of the nucleus
 - the radius of the nucleus is less than 10^{-14} m
 - scattering follows Coulomb's law
 - All of these
33. In a hydrogen atom sample, electrons are de-excited from 3rd excited state to ground state. Which of the following statement(s) is/are **correct**?
- No line is observed in Pfund series.
 - Total ten lines are observed in spectrum.
 - Four lines are observed in UV-region.
- I and III only
 - II and III only
 - I only
 - I, II and III
34. The ratio of 2nd Bohr orbit of Hydrogen atom to 1st orbit of Li^{2+} ion i.e., $r_2(\text{H}) : r_1(\text{Li}^{2+})$ is
- 12 : 1
 - 4 : 3
 - 4 : 1
 - 1 : 12
35. Mark the **incorrect** statement regarding the photoelectric effect.
- There is no time lag between the striking of light beam and the ejection of electrons from the metal surface.
 - The number of electrons ejected is inversely proportional to the intensity of light.
 - Photoelectric effect is not observed below threshold frequency.
 - The kinetic energy of the electrons increases with increase in frequency of light used.
36. For single electron species, the **correct** order of energy of the orbitals;
- $1s < 2s < 2p < 3s < 3p = 3d < 4s < 4d$
 - $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p$
 - $1s < 2s < 2p < 3s < 3p < 4s < 3d$
 - $1s < 2s < 2p < 3s < 3p < 3d < 4s$
37. The increasing order (lowest first) for the values of q/m (charge/mass) for:
- $e < p < n < \alpha$
 - $n < p < e < \alpha$
 - $n < p < \alpha < e$
 - $n < \alpha < p < e$
38. An electron in an atom is moving with a speed of 662 ms^{-1} with an accuracy of 0.001%. The uncertainty with which the position of electron in atom can be located is
(Given: $m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.62 \times 10^{-34} \text{ J s}$)
- $2.45 \times 10^{-4} \text{ m}$
 - $4.21 \times 10^{-3} \text{ m}$
 - $8.62 \times 10^{-4} \text{ m}$
 - $8.74 \times 10^{-3} \text{ m}$

39. The energy of I, II and III energy levels of a certain atom are E , $4E/3$ and $2E$ respectively. A photon of wavelength λ is emitted during a transition from III to I. What will be the wavelength of emission for transition II to I?
- $\lambda/2$
 - λ
 - 2λ
 - 3λ
40. If the energy gap between two levels is $\Delta E = 1.89 \text{ eV}$, what is the wavelength of the emitted photon? ($hc = 1240 \text{ eV} \cdot \text{nm}$)
- 121 nm
 - 164 nm
 - 486 nm
 - 656 nm

41. Match the **List-I** with **List-II**.

List-I (Quantum number for an electron in orbital)		List-II (Orbital)	
A	$n = 2, \ell = 1, m = 0, s = +\frac{1}{2}$	P	3d
B	$n = 3, \ell = 2, m = -2, s = -\frac{1}{2}$	Q	4f
C	$n = 1, \ell = 0, m = 0, s = +\frac{1}{2}$	R	2p
D	$n = 4, \ell = 3, m = -2, s = -\frac{1}{2}$	S	1s

Choose the **correct** answer from the options given below;

- A-Q; B-S; C-R; D-P
 - A-R; B-P; C-S; D-Q
 - A-P; B-R; C-Q; D-S
 - A-R; B-P; C-Q; D-S
42. A bulb of 40 W is producing a light of wavelength 620 nm with 80% of efficiency then the number of photons emitted by the bulb in 20 seconds are ($1\text{eV} = 1.6 \times 10^{-19} \text{ J}$) ($hc = 12400 \text{ eV} \cdot \text{\AA}$)
- 2×10^{18}
 - 10^{18}
 - 10^{21}
 - 2×10^{21}
43. If the number of paired electrons present in the subshell with $l = 2$, in Ni^{2+} and Cr^{2+} is x and y respectively, then the value of ' $x + y$ ' will be;
- 4
 - 3
 - 6
 - 1
44. If change in energy (ΔE) = $3 \times 10^{-8} \text{ J}$, $h = 6.6 \times 10^{-34} \text{ J-s}$ and $c = 3 \times 10^8 \text{ m/s}$, then wavelength of the light is;
- $6.36 \times 10^3 \text{ \AA}$
 - $6.36 \times 10^{-10} \text{ \AA}$
 - $6.6 \times 10^{-8} \text{ \AA}$
 - $6.36 \times 10^{18} \text{ \AA}$

45. Match List-I with List-II.

List-I (Type of nodes)		List-II (Number of node)	
A	The radial node of 5s atomic orbital	P	1
B	The angular node of $3d_{yz}$ atomic orbital	Q	4
C	The total node of $4d_{xy}$ atomic orbital	R	2
D	The angular node of 3p atomic orbital	S	3

Choose the **correct** answer from the options given below:

- (1) A-Q; B-R; C-P; D-S
 (2) A-Q; B-R; C-S; D-P
 (3) A-R; B-P; C-Q; D-S
 (4) A-P; B-R; C-Q; D-S

Integer Type Questions (46-50)

46. The orbital angular momentum of an electron in 3s orbital is $\frac{xh}{2\pi}$. The value of x is:

47. If the velocity of the electron in Bohr's first orbit is 2.19×10^6 m/s, calculate the de Broglie wavelength (in pm) associated with it. (Nearest integer)
48. The wavelength of the spectral line, when the electron in the hydrogen atom undergoes a transition from the energy level 4 to energy level 2 is _____ nm. (Nearest Integer)
 ($R_H = 109678 \text{ cm}^{-1}$)
49. The sum of number of electrons, protons and neutrons in $^{80}_{35}\text{Br}$ is ____
50. The maximum number of electrons that can have principal quantum number, $n = 3$, and spin quantum $m_s = -\frac{1}{2}$, is

SECTION-III (MATHEMATICS)

Single Correct Type Questions (51-70)

51. Solution set of equation $\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} = 3$ is

- (1) $\pm\sqrt{\frac{3}{5}}$ (2) ± 4
 (3) $\pm\frac{1}{\sqrt{2}}$ (4) ± 6

52. Complete solution set of inequality

$$\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1 \text{ is equal to}$$

- (1) $(-\infty, 0]$ (2) $(-\infty, 0] \cup (2, 3)$
 (3) $[2, 3]$ (4) $(-\infty, 2) \cup (3, \infty)$

53. The value of

$$\frac{(1.5)^3 + (4.7)^3 + (3.8)^3 - 3 \times 1.5 \times 4.7 \times 3.8}{(1.5)^2 + (4.7)^2 + (3.8)^2 - 1.5 \times 4.7 - 1.5 \times 3.8 - 4.7 \times 3.8} \text{ is}$$

- (1) 8 (2) 9
 (3) 10 (4) 11

54. Complete set of values of 'x' for which

$$\frac{x^3 - 7x^2 + 10x}{x^2 + 5x + 6} \leq 0$$

- (1) $(-\infty, -3) \cup (-2, 0] \cup [2, 5]$
 (2) $(-3, -2) \cup [0, 2] \cup [5, \infty)$
 (3) $(-\infty, -3) \cup (-2, 0) \cup (2, 5)$
 (4) $(-3, -2) \cup (0, 2) \cup (5, \infty)$

55. If $a + b + c = 0$ then $\frac{a^2}{x^{bc}} \cdot \frac{b^2}{x^{ac}} \cdot \frac{c^2}{x^{ab}}$ is equal to

- (1) x (2) x^2
 (3) 1 (4) x^3

56. Solution set of $\frac{x^4(x+1)^2(x-2)}{(x-3)^3(x+4)} > 0$ is

- (1) $x \in (-4, 2) \cup (4, \infty)$
 (2) $x \in (-4, -1) \cup (-1, 2) \cup (3, \infty)$
 (3) $x \in (-4, 0) \cup (0, 2) \cup (3, \infty)$
 (4) $x \in (-4, -1) \cup (-1, 0) \cup (0, 2) \cup (3, \infty)$

57. 1. Numbers which are not prime are composite.
 2. If a and b are twin primes, then they are prime numbers with absolute difference 2.
 3. If a and b are co-prime numbers, then HCF (a, b) = 1.
 4. The sum of a rational and an irrational number is always irrational.

Which of the above statements is/are correct?

- (1) 1, 2 and 3 only (2) 1 and 4 only
 (3) 2, 3 and 4 only (4) 1, 2, 3 and 4

58. If $\frac{4+3\sqrt{5}}{4-3\sqrt{5}} = a+b\sqrt{5}$, a, b are rational numbers, then $(a, b) =$

- (1) $\left(\frac{61}{29}, \frac{-24}{29}\right)$ (2) $\left(\frac{-61}{29}, \frac{24}{29}\right)$
 (3) $\left(\frac{61}{29}, \frac{24}{29}\right)$ (4) $\left(\frac{-61}{29}, \frac{-24}{29}\right)$

59. Number of integer values of x satisfying $-5 \leq x < 10$ and $0 \leq x \leq 15$ is

- (1) 10 (2) 11
(3) 12 (4) 13

60. If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$ then the value of x will be

- (1) -1 (2) 1
(3) 2 (4) 3

61. If $\frac{1}{4.103} = 0.2437$, then the value of $\frac{1}{0.0004103}$ is

- (1) 2437 (2) 24370
(3) 2.437 (4) 243700

62. If $\frac{14x}{x+1} - \frac{9x-30}{x-4} < 0$, then complete solution set of the inequality is

- (1) $(-1, 1) \cup (3, 6)$ (2) $(-1, 1) \cup (2, 6)$
(3) $(-1, 1) \cup (4, 6)$ (4) $(-1, 1) \cup (5, 6)$

63. Value of $\left(\frac{x^{\frac{2}{3}}y^{\frac{1}{2}}}{yx^{\frac{2}{3}}}\right)^6 \div \sqrt{\frac{xy^{-2}}{yx^{-1}}}$ is equal to—

- (1) x^2 (2) x
(3) x^3 (4) x^4

64. Let $a, b \in \mathbb{Q}$ such that $\frac{39\sqrt{2}-5}{3-\sqrt{2}} = a + b\sqrt{2}$, then which is incorrect?

- (1) $\sqrt{\frac{b}{a}}$ is a rational number
(2) b and a are coprime rational numbers
(3) $b - a$ is a composite number
(4) $\sqrt{a+b}$ is a rational number

65. If $a + 2b = 3$ and $ab = -5$, then find the value of $a^2 + 4b^2$

- (1) 9 (2) -11
(3) 29 (4) 27

66. Values of ' x ' for which $(x^2 - 3)(x + 1)$ is always negative, are

- (1) $x \in (-\infty, -\sqrt{3}) \cup (-1, \sqrt{3})$
(2) $x \in (-\sqrt{3}, \sqrt{3})$
(3) $x \in \mathbb{R}$
(4) $x \in (-\sqrt{3}, -1) \cup (\sqrt{3}, \infty)$

67. Find the largest number which divides 245 and 1029 leaving remainder 5 in each case.

- (1) 8 (2) 10
(3) 12 (4) 16

68. If $x^2 + 4y^2 + 9z^2 - 2x - 4y - 6z + 3 = 0$ then, $x + y + z$ is equal to

- (1) $1/6$
(2) $11/6$
(3) $13/16$
(4) Cannot be determined

69. If the solution of the inequality

$$1 < \frac{3x^2 - 7x + 8}{x^2 + 1} \leq 2, \text{ is } [\alpha, \beta] \text{ then mark incorrect}$$

option

- (1) α is least natural number
(2) $\beta - \alpha$ is a prime number
(3) $\beta + \alpha$ is a prime number
(4) $\beta + 3\alpha$ is a prime number

70. If $n > 0$ and exactly 15 integers satisfy $(x + 6)(x - 4)(x - 5)(2x - n) \leq 0$, then the least possible value of n is

- (1) 10 (2) 12
(3) 14 (4) 16

Integer Type Questions (71-75)

71. The number of integral solution of the inequality

$$\frac{x^2(x+1)^{10}(x-2)^4}{(x+4)^7(x+7)^5(x-5)^3} \geq 0 \text{ is (where } x < 5)$$

72. Suppose that $a + x^2 = 2006$, $b + x^2 = 2007$ and $c + x^2 = 2008$ and $abc = 3$. Find the value of

$$\frac{a}{bc} + \frac{b}{ca} + \frac{c}{ab} - \frac{1}{a} - \frac{1}{b} - \frac{1}{c}$$

73. Assume that a, b, c, d are positive integers such that $a^5 = b^4$, $c^3 = d^2$ and $c - a = 19$. Then $d - b$ is equal to

74. If k is the number of integral values of x

$$\text{satisfying } \frac{x(x-1)^2(x^2-7x+10)(2x-1)^2}{(x-1)^5(2x-3)^2} \leq 0,$$

then $k =$

75. If $a, b, c \in \mathbb{N}$, then the least positive value of $a^2 + b^2 + c^2 - ab - bc - ca$ is



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